Product Introduction

ZDH1012 is a power amplifier manufactured using gallium arsenide (GaAs) HBT technology. It operates from DC to 1000MHz. The ZDH1012 is designed for single-voltage supply from 3.0V to 9.0V.

ZDH1012 is packaged in an SOT-89 package, offering excellent reliability and cost-effectiveness.

Typical application scenarios

- 315MHz, 433MHz ISM data transmission
- Walkie-talkies
- Remote control of aircraft models
- Meter reading systems

Absolute Maximum Ratings

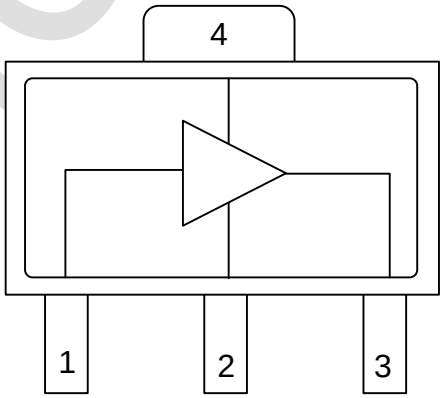
Parameter	Value
Storage Temperature	-65°C~+150°C
Operating Temperature	-55°C~+105°C
Supply Voltage (Vcc)	+12.0V
Input Power (RFIN)	+20.0dBm

Features

- 3.0V~9.0V single voltage supply, typical current: 335mA @ 9.0V, 388mA @ 5.0V
- Small signal gain: 22dB @ 433MHz, VCC=9V; 20dB @ 433MHz、VCC=5V
- Typical output power: 34dBm @ 433MHz, VCC=9V; 31dBm @ 433MHz、VCC=5V
- Efficiency: >65% @ 433MHz
- Input/output 50Ω impedance matching
- Green lead-free 3-pin SOT-89 package

This product complies with all relevant regulations and is halogen-free

Pin diagram (Top View)



RFIN GND RFOUT/VDD

PIN No.	Pin Name	Description
1	RFIN/VBIAS	RF Input
2	GND	Ground
3	RFOUT/VCC	RF Output/VCC

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Electrical Parameters

1. Test Conditions: Temp = +25°C, 433MHz application circuit, 50Ω test system.

Parameter	Unit	V_{CC} =9V	V_{CC} =7.4V	V_{CC} =5V	V_{CC} =3.3V
Gain	dB	22	21	21	21
Input Return Loss (S11)	dB	-24	-7	-7	-7
Output Return Loss (S22)	dB	-8	-8	-8	-9
Reverse isolation (S12)	dB	-42	-40	-40	-37
P1dB	dBm	34(1)	32(2)	29(3)	28(4)
Static current (Icq)	mA	290	330	400	400
Operating Current (Icc)	mA	390(1)	400(2)	409(3)	312(4)
Efficiency	%	64	65	66	61

Notes: (1) Input power is 15 dBm, VBIAS = 9V
(2) Input power is 15 dBm, VBIAS = 7.4V
(3) Input power is 15 dBm, VBIAS = 5V
(4) Input power is 12 dBm, VBIAS = 3.3V

2. Test conditions: VCC = +9V, Temp = +25°C, 30MHz~678MHz application circuit, 50Ω test system.

Parameter	Values			Unit
Frequency (f)	30	400	678	MHz
Gain	17	18	17	dB
Input Return Loss (S11)	-8	-6	-8	dB
Output Return Loss (S22)	-3	-8	-11	dB
P1dB	30	31	31	dBm
OIP3 (Tone=10dBm)	48	47	46	dBm
Static current (Icq)	285	285	285	mA
Operating Current (Icc)	280	335	345	mA
Efficiency	52	49	48	%

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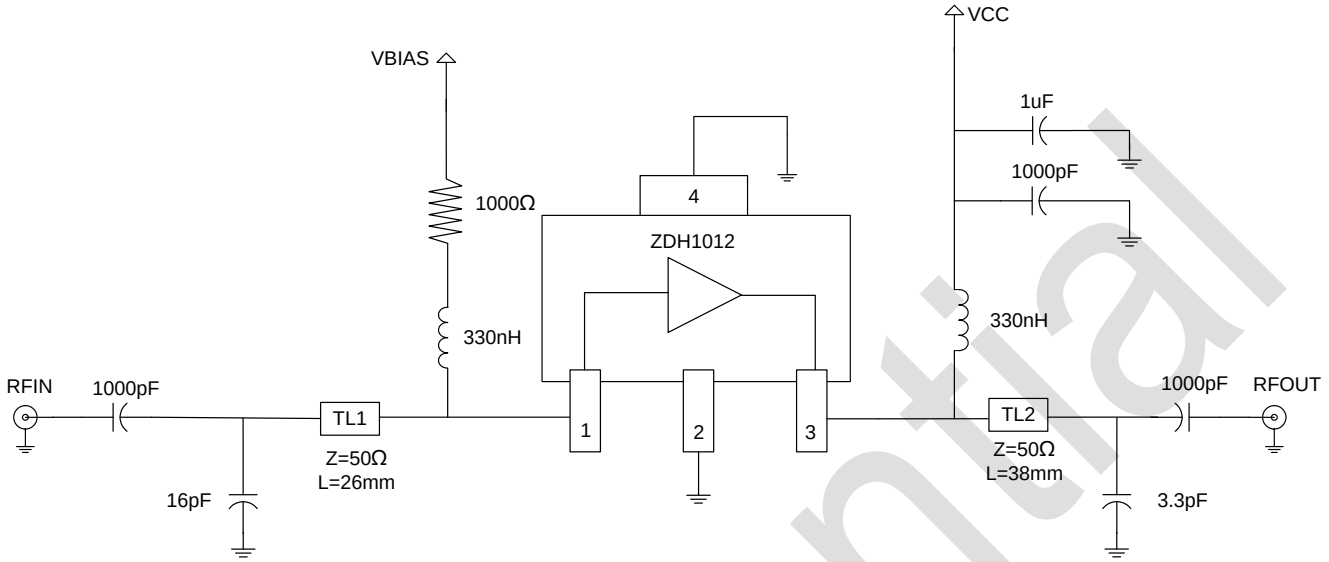
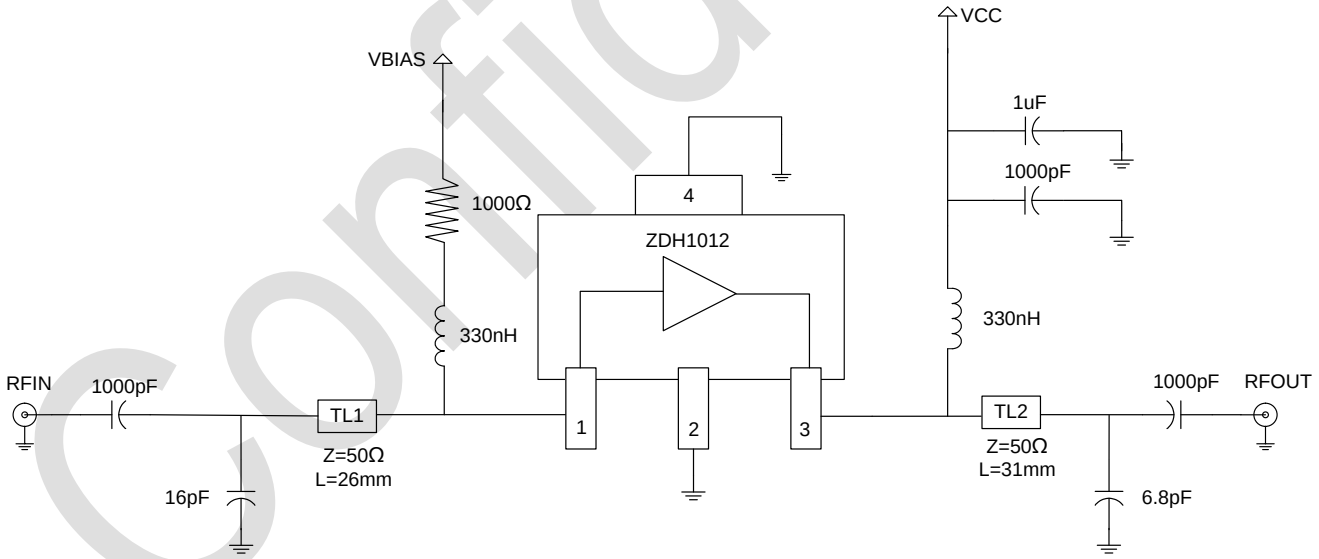
Typical application circuit: 433MHz

VCC=9V



VCC=5V



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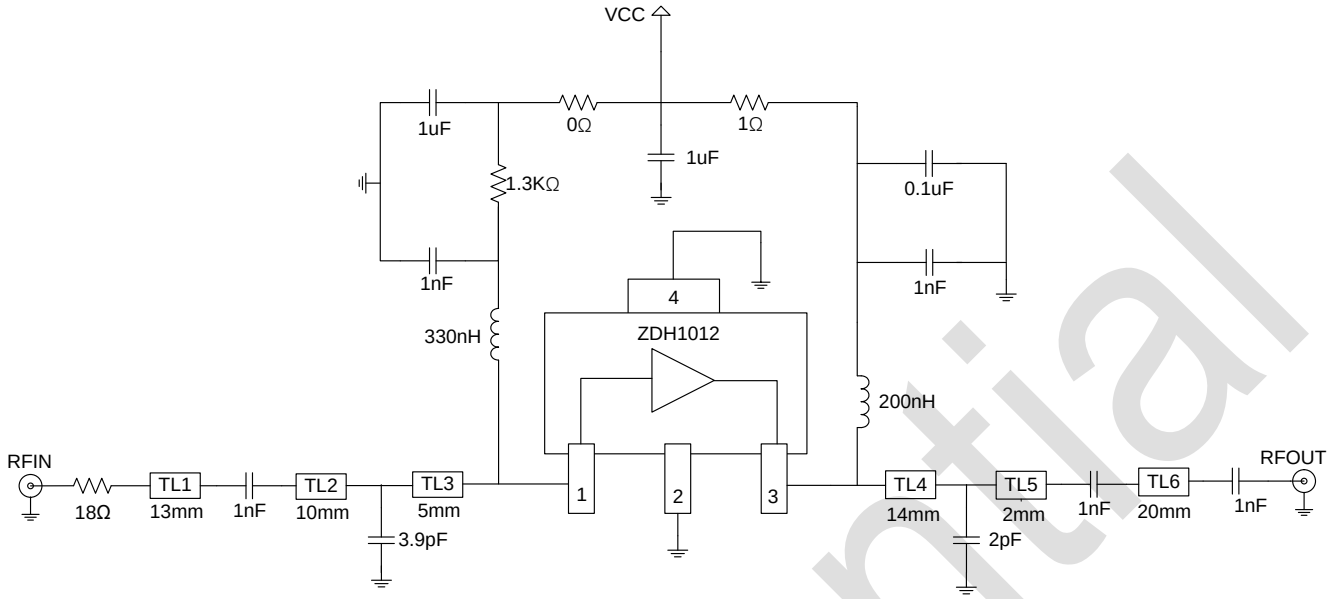
Typical application circuit: 30MHz~678MHz, 9V



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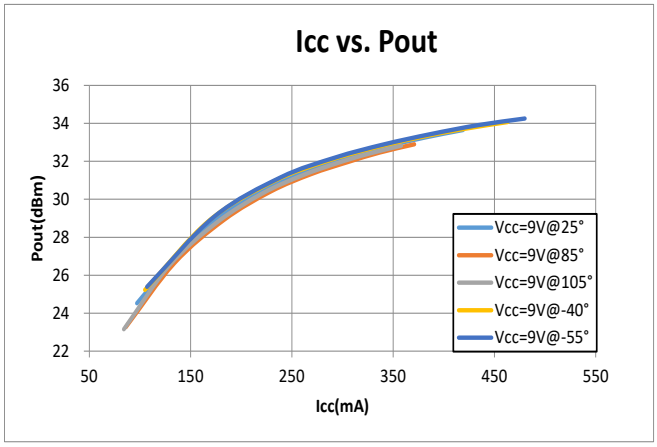
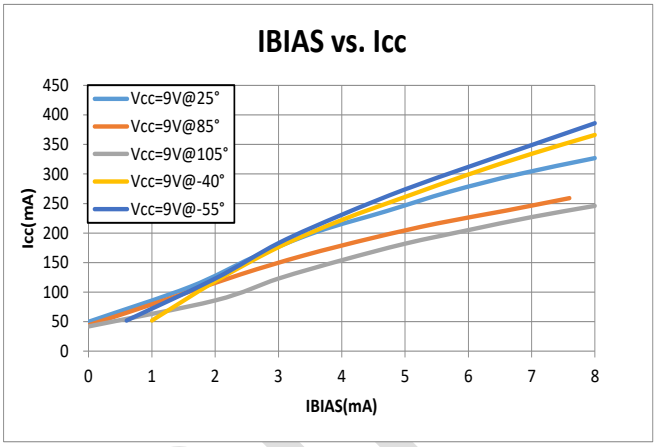
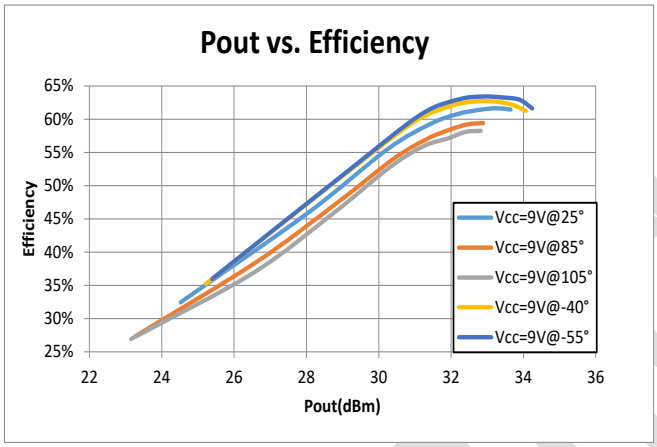
Typical performance curves

1. Test conditions: VCC=9V, 433MHz application circuit, 50Ω test system.







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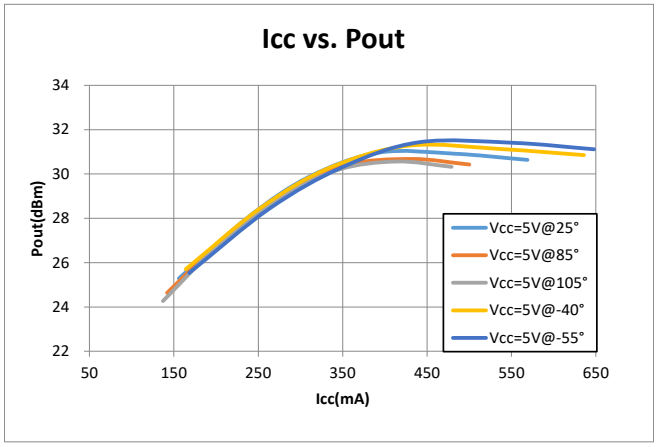
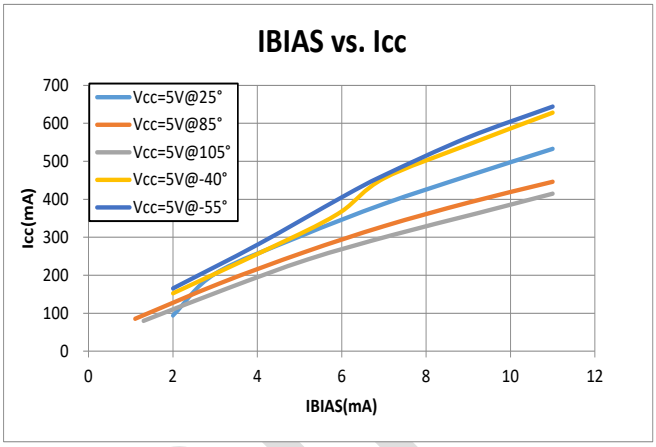
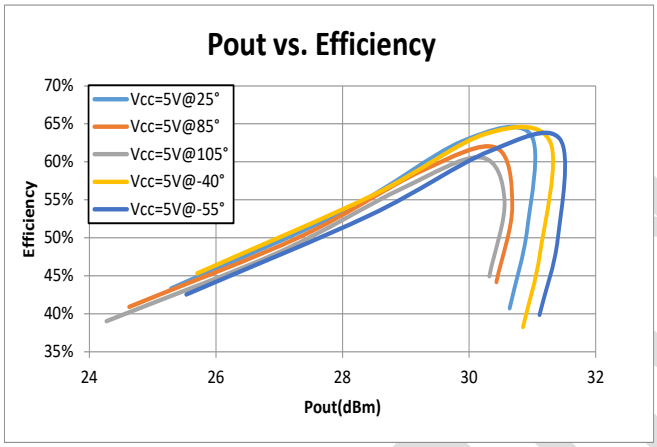
Typical performance curves

2. Test conditions: VCC=5V, 433MHz application circuit, 50Ω test system.







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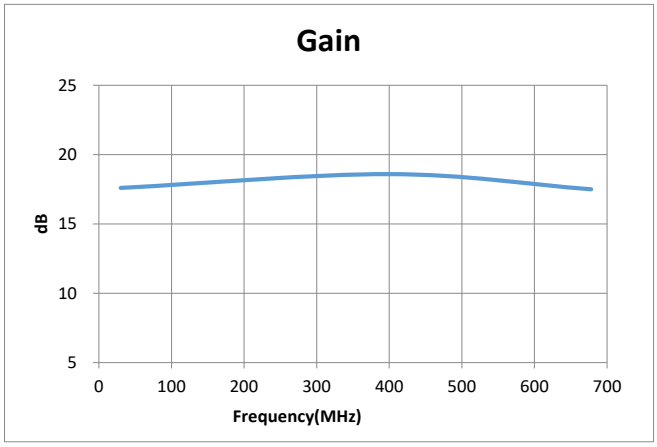
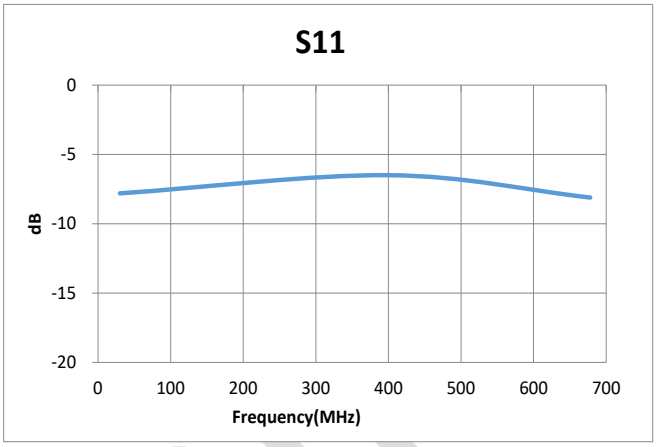
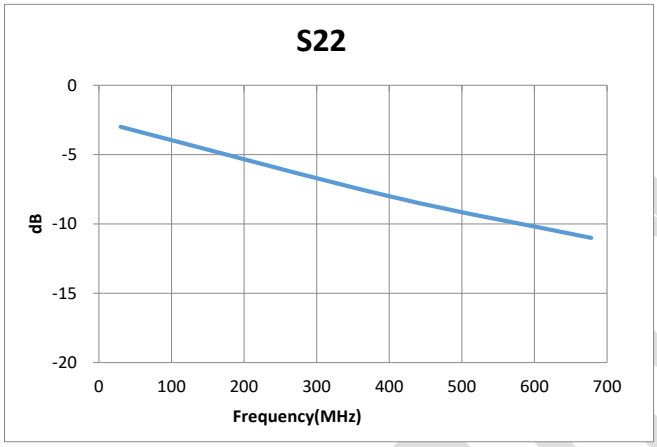
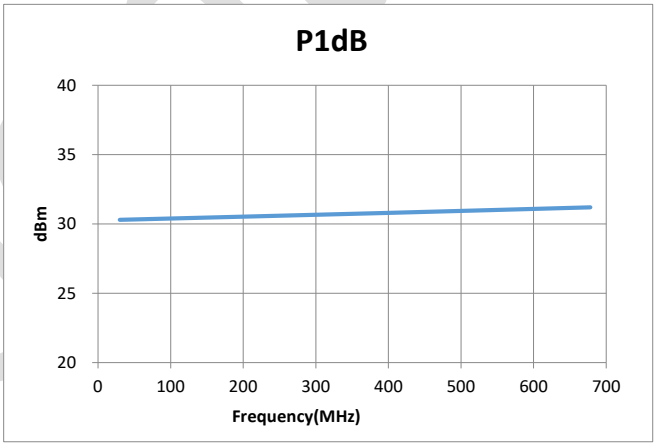
Typical performance curves

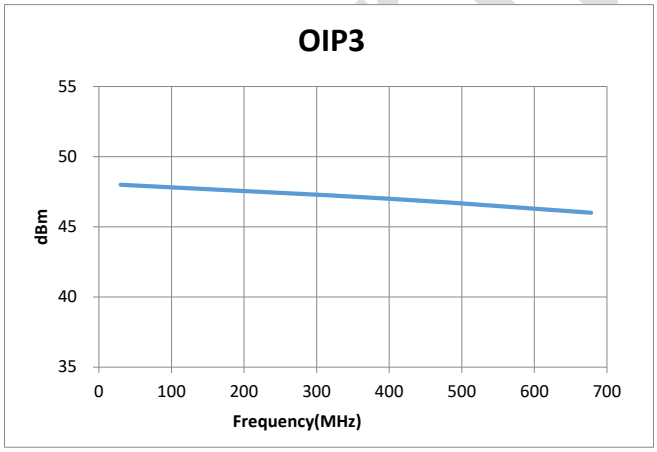
3. Test conditions: VCC=VBB=+9V, Temp=+25° C, 30MHz~678MHz application circuit, 50Ω test system.

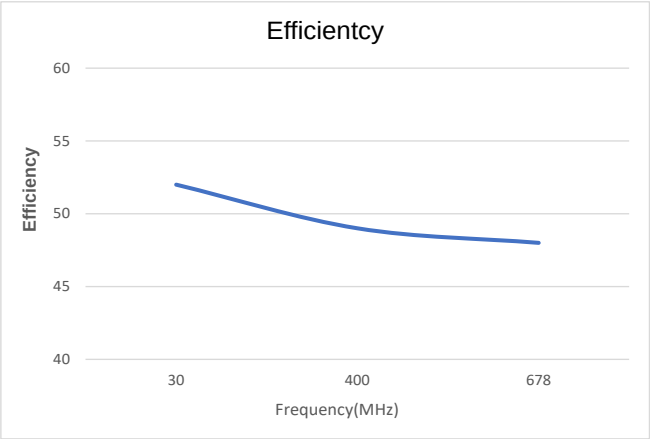




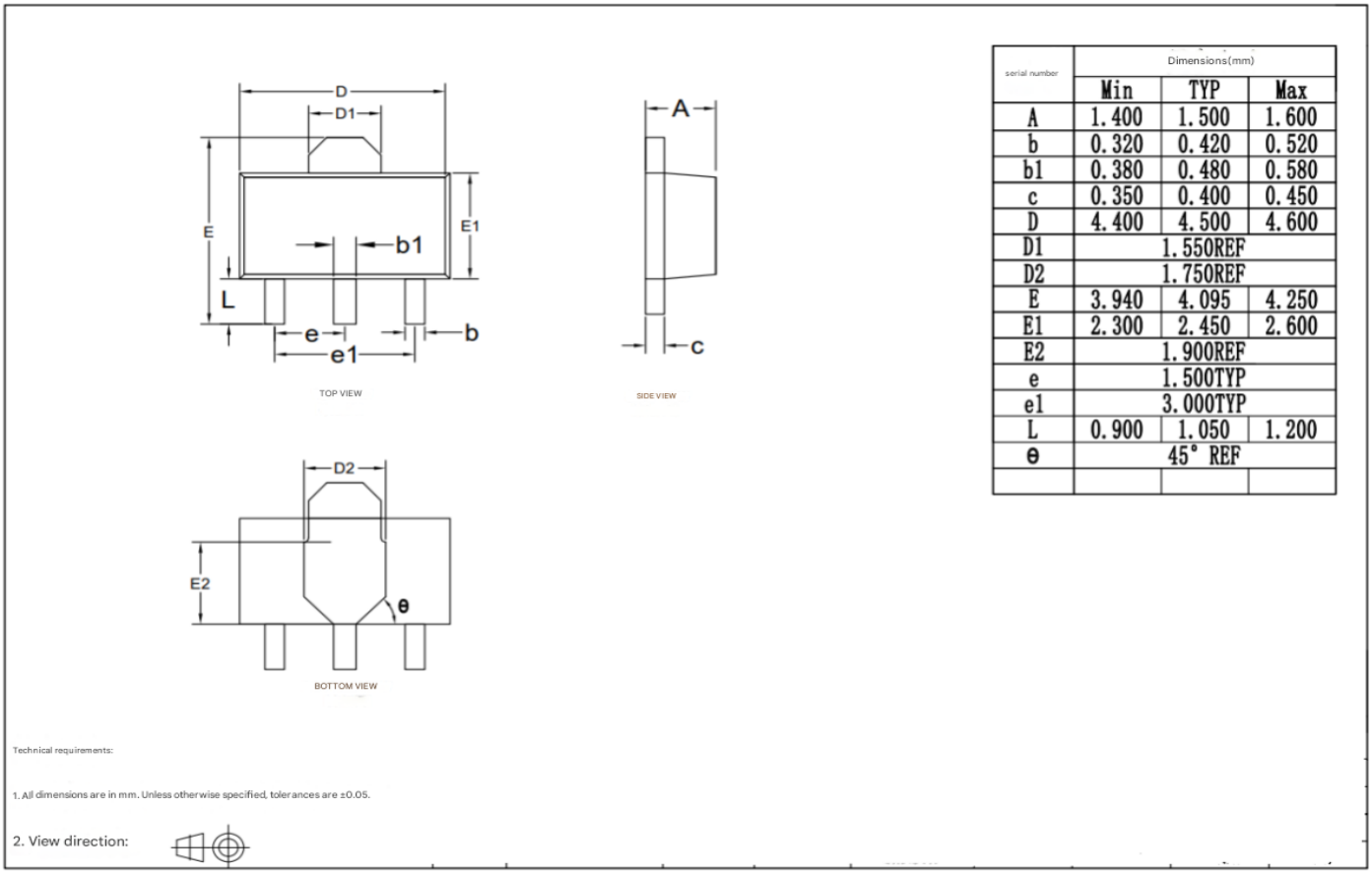








Package size diagram



serial number	Dimensions(mm)		
	Min	TYP	Max
A	1.400	1.500	1.600
b	0.320	0.420	0.520
b1	0.380	0.480	0.580
c	0.350	0.400	0.450
D	4.400	4.500	4.600
D1	1.550REF		
D2	1.750REF		
E	3.940	4.095	4.250
E1	2.300	2.450	2.600
E2	1.900REF		
e	1.500TYP		
e1	3.000TYP		
L	0.900	1.050	1.200
θ	45° REF		

Ordering Information

Model	Marking	Package
ZDH1012	1012T	SOT-89